1 The solar radiation incident on the surface of a shallow lake causes the temperature T of the lake to vary with time t. A climate model proposes that the rate of change of a lake's temperature is governed by the following equation:

$$\mu \frac{dT}{dt} = I_0 \cos(\omega t) - \kappa(T - T_G)$$

where,
I_0 is the peak power per unit area of the solar radiation on the lake,
T_G is the ground temperature, and
μ, ω, κ are constant coefficients.

What are the SI base units of the constant coefficient μ?

A $J\ m^{-2}\ K^{-1}$

B $kg\ s^{-3}\ K^{-1}$

C $J\ m^{-2}\ s^{-1}\ K^{-1}$

D $kg\ s^{-2}\ K^{-1}$